

Presentation Day Guide - Mini-Projet Big Data

This guide is for running the project from a fresh PC start until Docker shutdown.

1. Start the PC

1. Turn on the PC.
2. Open Docker Desktop.
3. Wait until Docker shows that it is running.
4. Open PowerShell.

2. Go to the Project Folder

```
cd "C:\Users\berqi\Documents\Mini-Projet-Big-Data"
```

Check that you are in the correct folder:

```
pwd
```

Expected folder:

```
C:\Users\berqi\Documents\Mini-Projet-Big-Data
```

3. Check the Dataset

The real Kaggle CSV must exist here:

```
data\Reviews.csv
```

Check it:

```
Test-Path data\Reviews.csv
```

Expected result:

```
True
```

If it returns False, copy Reviews.csv into the data folder before continuing.

4. Start All Services

Use this command:

```
docker compose up -d --build
```

Wait until the command finishes.

Check running services:

```
docker compose ps
```

Expected containers:

```
mnp_zookeeper
mnp_kafka
mnp_spark
mnp_spark_worker
mnp_airflow
mnp_api
```

5. Wait for Airflow

Airflow can take a little time to initialize.

Check logs:

```
docker logs --tail 60 mnp_airflow
```

Look for messages showing that the scheduler and webserver started.

Open Airflow:

```
http://localhost:8080
```

Login:

```
Username: admin
Password: admin
```

6. Check the API

Open in the browser:

```
http://localhost:8080/health
```

Expected result:

```
{"status": "ok"}
```

Or use PowerShell:

```
Invoke-RestMethod -Uri "http://localhost:8080/health"
```

7. Run the Airflow DAG

In Airflow:

1. Open the DAG named `recommendation\_pipeline`.

2. Unpause it if it is paused.
3. Click the trigger button.
4. Wait until all tasks become green.

Task order:

```
check_dataset
create_kafka_topic
stream_reviews_to_kafka
train_als_model
run_streaming_trigger
```

Alternative terminal test:

```
docker exec mnp_airflow airflow dags test recommendation_pipeline 2026-05-15
```

8. Verify Recommendations

Open:

```
http://localhost:8000/recommendations/user/A2R6RA8FRBS608
```

Expected result: a JSON response with a user id and recommended product ids.

Another useful user id:

```
A3NHUQ33CFH3VM
```

Open:

```
http://localhost:8000/recommendations/user/A3NHUQ33CFH3VM
```

9. Verify Recent Streaming Events

Open:

```
http://localhost:8000/events/recent?limit=5
```

Expected result: recent review events with:

```
user_id
product_id
score
time
```

10. Open the Dashboard

Open the dashboard file:

```
dashboard\index.html
```

Use it to:

1. Search recommendations by user id.
2. Refresh recent streaming events.

Recommended demo user:

```
A2R6RA8FRBS608
```

11. Useful URLs During the Demo

API health:
`http://localhost:8000/health`

Recommendations:
`http://localhost:8000/recommendations/user/A2R6RA8FRBS608`

Recent events:
`http://localhost:8000/events/recent?limit=5`

Airflow:
`http://localhost:8080`

Spark Master:
`http://localhost:8081`

12. Useful Debug Commands

Check all containers:

```
docker compose ps
```

Check API logs:

```
docker logs --tail 80 mnp_api
```

Check Airflow logs:

```
docker logs --tail 80 mnp_airflow
```

Check Kafka logs:

```
docker logs --tail 80 mnp_kafka
```

List Kafka topics:

```
docker exec mnp_kafka kafka-topics --bootstrap-server localhost:9092 --list
```

Read a few Kafka messages:

```
docker exec mnp_kafka kafka-console-consumer --bootstrap-server localhost:9092 --topic reviews --from-beginning
```

13. If Something Does Not Work

Docker is not running

Open Docker Desktop and wait until it is ready.

API health does not answer

Check containers:

```
docker compose ps
```

Restart API:

```
docker compose restart api
```

Airflow page does not open

Wait one minute, then check:

```
docker logs --tail 80 mnp_airflow
```

DAG fails because dataset is missing

Check:

```
Test-Path data\Reviews.csv
```

If missing, place Reviews.csv in:

```
data\Reviews.csv
```

Reset everything

Use this only if you want a clean restart:

```
docker compose down -v  
docker compose up -d --build
```

14. Shut Down Docker After the Demo

Stop the project:

```
docker compose down
```

Check that containers stopped:

```
docker compose ps
```

If you want to stop Docker Desktop too, close it after the containers are down.

15. Short Demo Script

1. Show GitHub repository and README.
2. Show Docker services running with `docker compose ps`.
3. Open Airflow at `http://localhost:8080`.
4. Explain the DAG tasks.
5. Trigger the DAG or show a successful run.
6. Open API health endpoint.
7. Open recommendations endpoint.
8. Open recent events endpoint.
9. Open dashboard and search a user.
10. Mention metrics:

```
Rows: 20000  
Distinct users: 17677  
Distinct products: 2657  
Validation RMSE: about 1.8974  
Test RMSE: about 1.9013
```

16. Final Shutdown

```
cd "C:\Users\berqi\Documents\Mini-Projet-Big-Data"  
docker compose down
```